THREE PERMANENCE OPERATIONS FOR CLOSED SUBSETS PROPER OVER A BASE: FOR F : X -> Y BETWEEN S-SCHEMES LOCALLY OF FINITE TYPE, IMAGES OF S-PROPER CLOSED SUBSETS ARE S-PROPER CLOSED WHEN Y IS SEPARATED OVER S OR F IS UNIVERSALLY CLOSED, AND INVERSE IMAGES OF S-PROPER CLOSED SUBSETS UNDER PROPER F ARE S-PROPER CLOSED.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7528-7541 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 7543-7588, the complete author proof (44 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33c. precard_gate RESULT=OK (probe L9999).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01W4: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W5: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T3: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W3: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03GN: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09MQ: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T5: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AH6: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JV: sources/stacks-coherent/pinned-original-source.tex, source lines 7588–7588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of schemes which are locally of finite type over S.*

1. *If Y is separated over S and $Z \subset X$ is a closed subset proper over S, then $f(Z)$ is a closed subset of Y proper over S.*
2. *If f is universally closed and $Z \subset X$ is a closed subset proper over S, then $f(Z)$ is a closed subset of Y proper over S.*
3. *If f is proper and $Z \subset Y$ is a closed subset proper over S, then $f^{-1}(Z)$ is a closed subset of X proper over S.*

Proof. Proof of (1). Assume Y is separated over S and $Z \subset X$ is a closed subset proper over S. Endow Z with the reduced induced closed subscheme structure and apply Morphisms, Lemma 0AH6 to $Z \to Y$ over S to conclude.

Proof of (2). Assume f is universally closed and $Z \subset X$ is a closed subset proper over S. Endow Z and $Z' = f(Z)$ with their reduced induced closed subscheme structures. We obtain an induced morphism $Z \to Z'$. Denote $Z'' = f^{-1}(Z')$ the scheme theoretic inverse image (Schemes, Definition 01JV). Then $Z'' \to Z'$ is universally closed as a base change of f (Morphisms, Lemma 01W4). Hence $Z \to Z'$ is universally closed as a composition of the closed immersion $Z \to Z''$ and $Z'' \to Z'$ (Morphisms, Lemmas 01W5 and 01W3). We conclude that $Z' \to S$ is separated by Morphisms, Lemma 09MQ. Since $Z \to S$ is quasi-compact and $Z \to Z'$ is surjective we see that $Z' \to S$ is quasi-compact. Since $Z' \to S$ is the composition of $Z' \to Y$ and $Y \to S$ we see that $Z' \to S$ is locally of finite type (Morphisms, Lemmas 01T5 and 01T3). Finally, since $Z \to S$ is universally closed, we see that the same thing is true for $Z' \to S$ by Morphisms, Lemma 03GN. This finishes the proof.

Proof of (3). Assume f is proper and $Z \subset Y$ is a closed subset proper over S. Endow Z with the reduced induced closed subscheme structure. Denote $Z' = f^{-1}(Z)$ the scheme theoretic inverse image (Schemes, Definition 01JV). Then $Z' \to Z$ is proper as a base change of f (Morphisms, Lemma 01W4). Whence $Z' \to S$ is proper as

the composition of $Z' \to Z$ and $Z \to S$ (Morphisms, Lemma 01W3). This finishes the proof. $\square$